Physical-Time Action Relaxation in an Elastic Collision Bath

navstokgap research working notes

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Abstract

An elastic repeated-collision model produces an action-valued covariance observable that relaxes exponentially in physical time to a positive constant. We derive its rate and plateau directly from the masses, bath velocity variance and collision frequency. Bounded bath velocities keep every collision below a fixed speed ceiling. The exact parameter dependence isolates the additional law needed for a universal action scale: energy and momentum conservation determine the collision map, while the reservoir statistics set the plateau.

1 A momentum receiver and a classical stochastic clock

A tracer of mass $m \geq M > 0$ meets fresh bath particles of mass M on the line. Collisions occur at an independent Poisson rate $\nu > 0$. Incoming bath velocities W_n are independent, identically distributed, centered and obey $|W_n| \leq w < c/3$, with variance $s^2 > 0$. Initial tracer velocity is independent of the bath and clock, centered, and bounded by w. Between collisions it is constant; $X(t) = X(0) + \int_0^t V(r)\,dr$.

The nontrivial one-dimensional elastic collision map is

$$V' = aV + bW, \quad W' = \frac{2m}{m+M}V - aW, \qquad a = \frac{m-M}{m+M}, \quad b = \frac{2M}{m+M} = 1 - a. \tag{1}$$

Direct substitution gives $mV' + MW' = mV + MW$ and $m(V')^2 + M(W')^2 = mV^2 + MW^2$. Since $0 \leq a < 1$, the tracer update is a convex combination and preserves $|V| \leq w$. The departing particle obeys $|W'| \leq (3m - M)w/(m + M) < 3w < c$.

The model specifies a refreshed reservoir and a velocity-independent clock. These are Barkai's collision-model premises [2, pp. 3–5, (2)–(12)], specialized here to a bounded bath and $m \geq M$. Each collision accounts for momentum and energy transfer; the complete stream supplies fresh particles and removes departing ones. A spatial gas realization would additionally derive encounter rates and incoming velocity statistics. The speed ceiling here is a kinematic restriction on Newtonian collisions in the reservoir frame, rather than a Lorentz-covariant collision law.

2 Stationary covariance and the action plateau

Define the two inverse-time rates

$$\gamma = \nu(1-a) = \frac{2\nu M}{m+M}, \qquad \beta = \nu(1-a^2) = \frac{4\nu mM}{(m+M)^2}. \tag{2}$$

The velocity generator acts on bounded functions as $Lf(v) = \nu\{\mathbb{E}[f(av + bW)] - f(v)\}$.

Proposition 1 (Collision plateau)**.** *The stationary velocity law is unique on* $[-w, w]$ *and is represented by the convergent series* $V_* \overset{d}{=} b \sum_{j=0}^{\infty} a^j W_j$*. Its variance and autocovariance are*

$$S_* := \operatorname{Var} V_* = \frac{b^2 s^2}{1-a^2} = \frac{M}{m} s^2, \qquad C(r) = S_* e^{-\gamma r}. \tag{3}$$

Thus the stationary displacement observable $\mathsf{h}(\Delta) = m \operatorname{Var}[X(t+\Delta) - X(t)]/\Delta$ *satisfies*

$$\mathsf{h}(\Delta) = H_* \left[1 - \frac{1-e^{-\gamma\Delta}}{\gamma\Delta}\right], \qquad H_* := \frac{2mS_*}{\gamma} = \frac{(m+M)s^2}{\nu} > 0. \tag{4}$$

Proof. The series converges absolutely and stays in $[-w, w]$, since $b \sum a^j = 1$ (for $a = 0$ it is simply W_0). Separating its first term proves invariance under one collision. Coupling two initial velocities to the same clock and bath multiplies their difference by a^{N_t}. Its mean contraction factor is $\mathbb{E} a^{N_t} = e^{-\nu(1-a)t}$, also for $a = 0$ with $0^0 = 1$. Hence invariant probability laws on the bounded interval coincide. Independence and centering give the variance in (3). Conditioning on the collision count yields $\mathbb{E}[V(t+r) \mid V(t)] = e^{-\gamma r} V(t)$; stationary covariance follows. Integrating the covariance over the displacement square gives $\mathsf{h}(\Delta) = 2m \int_0^{\Delta} (1 - r/\Delta) C(r)\, dr$, which evaluates to (4). □

This proof uses a contracting affine collision update; it requires neither a finite velocity state space nor detailed balance of the refreshed bath. The energy-valued reservoir parameter $\Theta := Ms^2$ gives $mS_* = \Theta$ and $H_* = \Theta(m+M)/(\nu M)$. For a general bounded bath, Θ denotes twice the mean incoming kinetic energy, without assuming a thermodynamic equilibrium distribution. The stationary covariance and Green–Kubo integral also follow from Barbier and Trizac's field-free Maxwell-kernel specialization [1, p. 19, (89)–(98)]. Their kernel exponent called ν is zero there; our ν is a positive collision frequency.

3 An observable relaxing in physical time

The integrated future-lag covariance defines a second operational quantity:

$$\mathsf{a}(t) := 2m \int_0^{\infty} \operatorname{Cov}[V(t+r), V(t)]\, dr. \tag{5}$$

It has action units and uses physical preparation time t, with the lag integrated out. Its definition differs from the finite-duration displacement observable; both share the same stationary long-duration plateau.

Proposition 2 (Physical-time relaxation)**.** *For the centered initial ensemble specified above, with* $S_0 = \operatorname{Var} V(0)$*,*

$$S(t) = S_* + (S_0 - S_*) e^{-\beta t}, \tag{6}$$

$$\mathsf{a}(t) = \frac{2mS(t)}{\gamma} = H_* + (\mathsf{a}(0) - H_*) e^{-\beta t}, \qquad \dot{\mathsf{a}} = \beta(H_* - \mathsf{a}). \tag{7}$$

In particular a tracer prepared at rest has $\mathsf{a}(t) = H_*(1 - e^{-\beta t}) > 0$ *for every* $t > 0$ *and* $\mathsf{a}(t) \to H_* > 0$ *as* $t \to \infty$*.*

Proof. The generator gives $Lv = -\gamma v$ and $L(v^2) = -\beta v^2 + \nu b^2 s^2$. Centering is preserved, so $\dot{S} = -\beta S + \nu b^2 s^2 = \beta(S_* - S)$. The conditional mean from Proposition 1 gives, even away from stationarity, $\operatorname{Cov}[V(t+r), V(t)] = e^{-\gamma r} S(t)$. Integration proves (7). □

The original displacement definition retains a distinct duration parameter:

$$\mathsf{h}_\Delta(t) = \frac{2m}{\gamma\Delta}\int_0^\Delta S(t+r)\left[1-e^{-\gamma(\Delta-r)}\right]dr. \tag{8}$$

Indeed, for $r \le s$ the covariance inside its double integral is $e^{-\gamma(s-r)}S(t+r)$; integrating over s proves the formula. At fixed Δ, it approaches the stationary value in (4) exponentially at rate β. At every fixed t it tends to zero as $\Delta \downarrow 0$, with $\mathsf{h}_\Delta(t) \sim mS(t)\Delta$ when $S(t) > 0$. Its long-duration limit is H_*, since the transient contribution to (8) is bounded in magnitude by $2m|S_0 - S_*|e^{-\beta t}/(\gamma\beta\Delta)$.

4 What would select a common constant?

For fixed bath mass, variance and clock, the plateau grows as $m + M$. Achieving the same H_* for different tracer masses requires

$$\nu_m = \frac{(m+M)s_m^2}{H_*}. \tag{9}$$

This equation is a diagnostic of a proposed encounter law. Binary energy and momentum conservation alone place no restriction of this form on the external clock or reservoir preparation.

More explicitly, take a fixed admissible bath and a tracer initially at rest. For any $0 < \epsilon \le 1$, replace every incoming velocity by $W_n^{(\epsilon)} = \epsilon W_n$, keeping masses and clock unchanged. Every elastic collision, centering assumption and speed bound remains valid. Both H_* and $\mathsf{a}(t)$ are multiplied by ϵ^2. Consequently this entire class admits arbitrarily small positive plateaus even though every nondegenerate member relaxes to its own positive constant. This is a countermodel test for these specified stochastic collision premises.

The next model should replace the supplied clock by a spatial encounter law and test how it scales with reservoir velocity and density. The physical-time relaxation mechanism is already explicit: persistent injection of velocity variance balances collisional contraction. A universal quantum action scale requires an additional physical relation that fixes this balance across systems.

Ben-Naim and Krapivsky provide a useful clock comparison: exponential decay in collision number becomes algebraic decay in physical time in their cooling model [3, p. 6, (14)–(17)].

5 Return to the partition question

The project's central question concerns a sequence of time partitions $\pi_N = \{0 = t_0^{(N)} < \cdots < t_N^{(N)} = T\}$ with mesh tending to zero. The present relaxation law describes preparation time under a supplied bath. It therefore serves as a comparison mechanism for the partition problem: one must derive, from admissible cut-point refinements, both the quantity that survives and the condition selecting its positive value. The collision calculation isolates how a positive constant can instead enter through reservoir energy and a clock. The next partition test tracks these inputs explicitly when a refinement law is proposed.

References

[1] Matthieu Barbier and Emmanuel Trizac. Field induced stationary state for an accelerated tracer in a bath. arXiv:1203.6759v3, 2012. Selected pp. 1, 19; (89)–(98).

[2] Eli Barkai. Stable equilibrium based on Lévy statistics: A linear Boltzmann equation approach. arXiv:cond-mat/0303255v1, 2003. Selected pp. 1, 3–5.

[3] E. Ben-Naim and P. L. Krapivsky. The inelastic Maxwell model. arXiv:cond-mat/0301238v1, 2003. Selected pp. 1, 6; (14)–(17).